(a) Tasks and temporal functions

Selected examples: blue = GTS orange = TSK

	Forecasting and prediction	Augmentation and synthesis	Anomaly detection and diagnosis	Decision support
Data management	History and known covariates	Source series and generation conditions	Sensor streams and operating records SAGE-AD · GTS	Observations and action history
Temporal processing	Windows and multiscale features TimeSeriesScientist · GTS	Segments and transformations	Local changes and channel relations	Decision windows and state summaries
Context and memory	Events and historical cases	Conditions and reference examples MERIT · TSK	Incident context and fault history	Past actions and market information FinMem · TSK
Temporal validation	Information cutoff and time alignment	Temporal structure and condition fidelity	Temporal location and evidence	Evidence available at the action time

(b) Illustrative path: demand forecasting and inventory decisions

General execution	Temporal preparation	Forecast procedure	Inventory decision
Run tools; retain artifacts	Align data; retrieve context	Select and revise forecasts	Choose actions under costs
Artifact test: TimeSeriesGym	Context test: Dr-CiK	Model resources: ModernTSF	Scenario test: inventory simulator